MATHS MANTHAN
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Subject: Mathematics | **PRE-BOARD MOCK EXAMINATION – 12 (QUADRATIC EQUATIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 13(iii).
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions (VSAQ) of 2 marks each.
4. **Section C** comprises 2 Short Answer questions (SA) of 3 marks each.
5. **Section D** comprises 2 Long Answer questions (LA) of 5 marks each.
6. **Section E** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A (Multiple Choice & Assertion-Reason Questions 6 × 1 = 6 Marks)

Q1. The value of $\sqrt{6+\sqrt{6+\sqrt{6+\dots}}}$ is: [1]
(A) 4 (B) 3 (C) -2 (D) 3.5

Q2. If the quadratic equation $x^2 - kx + 4 = 0$ has no real roots, then: [1]
(A) $-4 < k < 4$ (B) $k > 4$ (C) $k < -4$ (D) $k = \pm 4$

Q3. If the sum and product of the roots of the equation $kx^2 + 6x + 4k = 0$ are equal, then $k =$: [1]
(A) $-\frac{3}{2}$ (B) $\frac{3}{2}$ (C) $\frac{2}{3}$ (D) $-\frac{2}{3}$

Q4. If one root of the quadratic equation $2x^2 + kx + 4 = 0$ is 2, then the other root is: [1]
(A) 6 (B) -6 (C) -1 (D) 1

Q5. If the roots of the quadratic equation $x^2 - px + q = 0$ differ by 1, then: [1]
(A) $p^2 = 4q + 1$ (B) $p^2 = 4q - 1$ (C) $q^2 = 4p + 1$ (D) $p^2 + 4q = 1$

Q6. Assertion-Reason: [1]
Directions: (A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.
Assertion (A): The quadratic equation $(x-2)^2 = 0$ has two equal real roots: $x = 2, 2$.
Reason (R): A quadratic equation always possesses exactly two roots (which may be distinct, coincident, or non-real).

SECTION B (Very Short Answer Questions 2 × 2 = 4 Marks)

Q7. Solve the following equation for x: [2]

$$\frac{1}{x} - \frac{1}{x-2} = 3, \quad x \neq 0, 2$$

Q8. Find the discriminant of the quadratic equation $3x^2 - 2x + \frac{1}{3} = 0$, and hence describe the nature of its roots. [2]

SECTION C (Short Answer Questions $2 \times 3 = 6$ Marks)

Q9. Solve the following quadratic equation for x by factorization: [3]

$$\sqrt{2}x^2 + 9x + 4\sqrt{2} = 0$$

Q10. Solve the following algebraic equation for x: [3]

$$\frac{2x}{x-3} + \frac{1}{2x+3} + \frac{3x+9}{(x-3)(2x+3)} = 0, \quad x \neq 3, -\frac{3}{2}$$

SECTION D (Long Answer / HOTS Questions $2 \times 5 = 10$ Marks)

Q11. Solve the following algebraic equation for x: [5]

$$\frac{1}{x+1} + \frac{2}{x+2} = \frac{4}{x+4}, \quad x \neq -1, -2, -4$$

Q12. A shopkeeper buys a number of books for Rs. 80. If he had bought 4 more books for the same total expenditure, each book would have cost Rs. 1 less. Determine how many books he originally bought and the original cost of each book. [5]

SECTION E (Applied Case Study Problem $1 \times 4 = 4$ Marks)

Q13. Case Study: School Educational Excursion & Group Budget Dynamics

The student council of a senior secondary school planned a leadership excursion to a science exploration centre with a fixed bus charter contract of Rs. 2000, to be shared equally among all participating students. On the morning of the departure, 5 students fell ill and could not join. Consequently, the remaining students each had to contribute Rs. 20 more to cover the total bus charter expense.

(i) If x denotes the original planned number of participating students, write expressions for the original and revised individual contributions. [1]

(ii) Formulate the quadratic equation in x in standard form representing the financial situation. [1]

(iii) Determine the original number of students planned (x) and the number of students who actually attended the excursion. [2]

OR

Calculate the exact amount each attending student paid and the extra burden incurred per student due to the 5 cancellations.

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